We thank the referee for their careful reading and constructive comments, which have helped us improve the clarity and interpretation of our results. Below, we respond to each comment in turn and describe the corresponding revisions to the manuscript (highlighted in red).

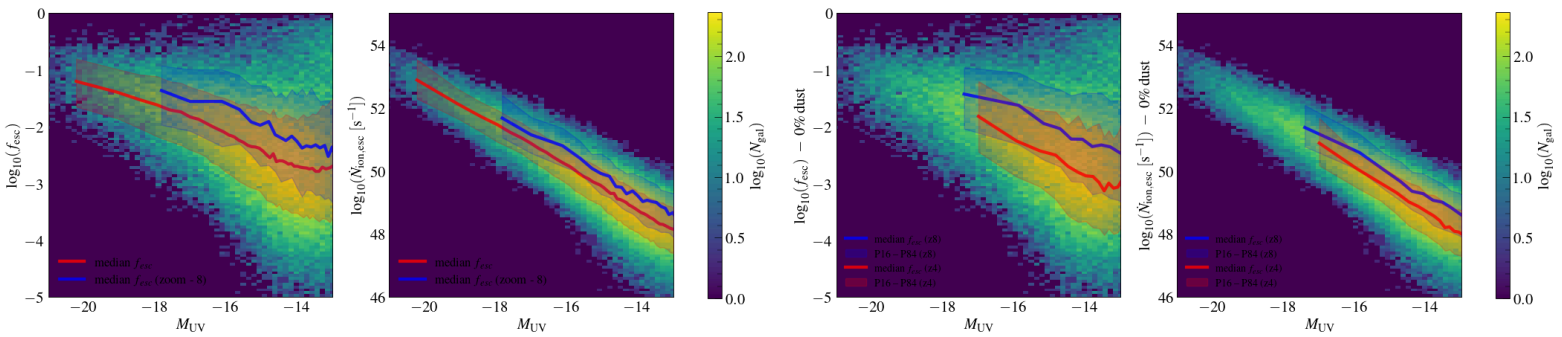
- The authors analyse one zoom realisation at the lowest resolution tier of the THESAN-ZOOM suite. Has any work been done to ascertain the convergence of LyC escape fractions/associated trends across these resolution tiers?

We thank the referee for raising this point. To investigate the convergence of LyC escape fractions and associated trends across resolution tiers, we have reproduced Fig. 1 of our paper twice below, comparing galaxy samples from both the 4x and 8x resolution tiers.

The left figure shows the median trends and percentile ranges for the full available 4x and 8x galaxy samples. For the right figure, we select samples matched in stellar mass and redshift to minimise differences arising from the underlying galaxy populations, particularly given the substantially larger size of the available 4x sample, and recalculate the median trends and percentile ranges accordingly. In both figures, the trends are overlaid on the binned distribution of the combined galaxy population.

We find that the resolution tiers exhibit qualitatively similar trends in both LyC escape fraction and ionizing photon escape rate with UV luminosity. However, a clear systematic offset in the normalisation of these relations remains, including after matching the samples in stellar mass and redshift. We therefore cannot claim numerical convergence of the normalisation of these relations. The limited number of galaxies available at 16x resolution prevents a robust assessment at this tier.

We retain the 4x sample for its broader coverage and to ensure a uniform numerical resolution throughout our analysis. We acknowledge that this does not eliminate potential systematic resolution effects on the quantitative calibration of the inferred relations and the resulting reionization histories, and have added a brief discussion of this limitation to Section 2.1.



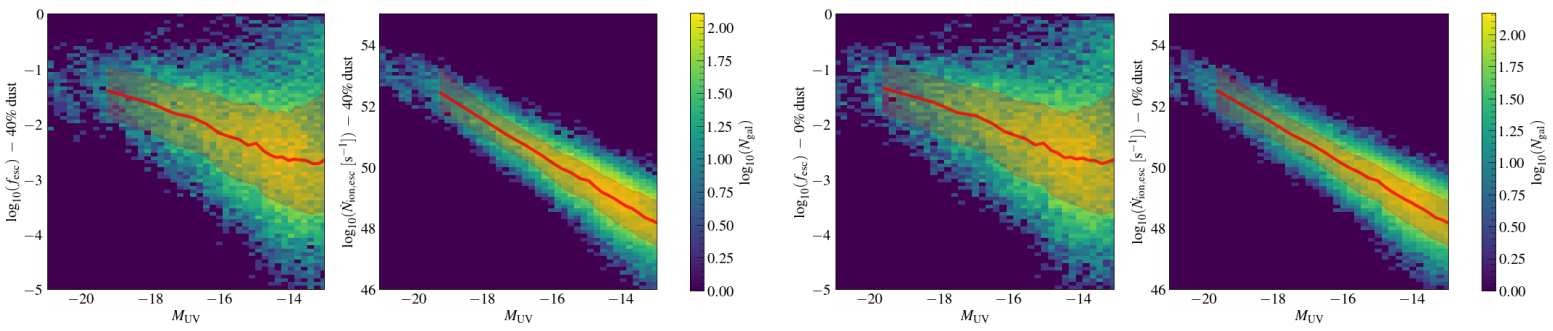
- For the COLT post-processing of LyC escape fractions, what dust attenuation shape, albedo, and opacity treatments were used?

Dust absorption and scattering in COLT use the wavelength-dependent cross-sections, albedos, and scattering asymmetry parameters of the Weingartner & Draine (2001) grain models (Milky Way RV = 3.1 size distributions). Rather than a single mixed-grain model, these are applied separately to silicate, graphite, and PAH grains. The dust masses in each cell are taken from the live dust model with the metal boost (see below), which follows dust production, growth, and destruction on the fly (McKinnon et al. 2017; Kannan et al. 2025), and 1 per cent of the carbonaceous mass is assigned to PAHs. Scattering is anisotropic and the attenuation curve emerges from the radiative transfer and depends on the local grain composition and on the relative geometry of stars and dust in each galaxy. We have added these details to Section 2.2.

- I understand the reasoning behind the need for a boost to the dust attenuation (page 3 line 51R: “both the live dust model and an additional contribution of 40% of the metals to dust”). How was this done in practice? Is this simply a boost to the DTM ratio? Have you tested the impact of this attenuation boost on measured LyC escape fractions?

The additional dust is added in each cell as a fixed fraction, fdust = 0.4, of the metal mass, on top of the live dust. It is divided between silicate and graphite grains in proportion to their live abundances:
DS' = DS + fdust x Z x DS / D, DG' = DG + fdust x Z x DG / D,
where DS and DG are the silicate and graphite dust-to-gas ratios from the live dust model, D = DS + DG, and Z is the gas metallicity. The total dust-to-gas ratio therefore becomes D + 0.4Z. This is equivalent to raising the dust-to-metal ratio by 0.4 above its live value while preserving the live grain composition.

We have tested the impact on LyC escape by repeating the COLT calculation with the live dust alone (see figure below; Left: 40% metal contribution to dust; Right: 0% metal contribution to dust). The additional dust lowers the median f_esc from 5.79 x 10^-3 to 5.66 x 10^-3 and the median N_ion,esc from 1.439 x 10^49 s^-1 to 1.430 x 10^49 s^-1, negligible compared with the 0.83 and 0.63 dex scatter observed in these quantities. We therefore see that the attenuation boost has very minimal impact on the LyC escape fractions and escape rates. This is consistent with previous studies which find that LyC absorption by dust is negligible due to an optical depth which is always much smaller than photoioization (HI; Yoo+2020, Mauerhofer+2021; Rosdahl+2022). The attenuation boost has been adopted here primarily to make the 1500Å UV Attenuation values realistic rather than to modify the measured LyC escape.



- How many photon packets are used per halo for the COLT MCRT?

COLT is run once per snapshot on the full high-resolution region, rather than separately for each halo. Each run uses ≈ (1–2) × 10^9 photon packets, emitted from all star particles in proportion to their ionizing luminosity, so the number of packets per galaxy scales with its intrinsic ionizing output. We have added this information to Section 2.2.

- Is the escape fraction discussed here the global (angle-averaged) value or the line-of-sight escape fraction?

The escape fraction considered here is the angle-averaged, global value. We have clarified this in Section 2.2.

- What ionizing SEDs were used to seed the photon packets?

The photon packets are seeded with the BPASS v2.2.1 binary stellar population models (Eldridge et al. 2017; Stanway & Eldridge 2018) for a Chabrier (2003) IMF with an upper mass limit of 100 Msun, interpolated in age and metallicity for each star particle. These are the same models that source the on-the-fly radiative transfer in THESAN-ZOOM (Kannan et al. 2025). We have added this to Section 2.2.

- Are the escape fractions discussed monochromatic (i.e. 900A-912A), or one including photons out to the helium edge (504A-912A)?

The escape fractions and escape rates are integrated over all hydrogen-ionizing photons (E > 13.6 eV). COLT samples the ionizing spectrum with 31 frequency bins between 1 and 912 A, with bin edges at the HI, HeI and HeII ionization thresholds (912, 504 and 228 A) and additional logarithmically spaced bins in between. fesc is therefore the photon-number-weighted escape fraction over the full ionizing spectrum, fesc = Nion,esc/Nion, which is the quantity required for the ionizing emissivity in Eq. (1). It can differ from the monochromatic escape fraction just below 912 A measured in observations, and we now note this in Section 2.2.

- Is the nebular continuum included in the Muv computation? This is a feature that will modify Muv and is directly dependent on the LyC escape fraction.

Nebular continuum emission is not included in the calculation of M_uv. We have clarified this in Section 2.2.

- Figure 1 caption: “brighter galaxies exhibit higher escape fractions”. This is true for the running median, but the highest escape fractions actually appear for the faintest sources. Please make this clear.

We thank the referee for pointing this out and have clarified the caption accordingly. The high escape fractions of some faint galaxies reflect the increased scatter and larger sample size at lower luminosities, rather than the typical trend. Moreover, their low intrinsic luminosities limit their contribution to the ionizing photon budget, while Thesan-Zoom resolution effects become increasingly important towards the faint end, where we impose a stellar-mass completeness limit of M_* ~ 10^6 M_☉. We clarify this in Section 4.1, but, acknowledging the referee's point, have softened our discussion of the f_esc - M_uv relation in Section 3.1 and modified the caption.

- Given the importance ultimately played by the relation between fesc and Muv, the authors must comment on and explain why their Muv trend disagrees with both the LzLCS as well as the described trend from Rosdahl+2022. Similarly, it would be good to include a broader literature discussion, including observational (Steidel+2018;Pahl+2021;Saxena+2024;Giovinazzo+2026;Jecmen+2026) and simulation (Anderson+2017;Naidu+2020;Yeh+2023;Choustikov+2024b) discussions of this trend. For example, Choustikov+2024b used a more detailed analysis of SPHINX to show that a strong trend between Muv and fesc only appears at fixed stellar mass. Similarly, Yeh+2023 used THESAN (a precursor to THESAN-ZOOM) to show a complex, redshift-dependent relationship between fesc and SFR (well-traced by Muv). Moreover, the phrase “high UV luminosity correlates with a high escape fraction” (page 4 line 47L) is incorrect. In fact, the very highest escape fractions only exist amongst the UV-faint population. How impacted are these results by a smaller number of galaxy realisations at the high-mass end?

We thank the referee for this important point. We have expanded our discussion of the relationship between f_esc and M_uv and its comparison with both observational and simulation-based studies. We agree that the diversity of trends reported in the literature warrants greater emphasis. Importantly, LzLCS is a low-redshift sample and is unlikely to be representative of the high-redshift population considered here. Furthermore, the THESAN simulations employ substantially different galaxy-formation models from THESAN-ZOOM, while the explicit modelling of the multiphase ISM in THESAN-ZOOM enables a more detailed and meaningful treatment of LyC escape. SPHINX provides a closer comparison and exhibits a broadly similar increase in f_esc towards brighter galaxies over much of the luminosity range, although with a turnover at the very bright end. This variation at high luminosities may relate to the tendency of galaxies in SPHINX to form more ordered disks with smoother star-formation histories at higher masses when compared to THESAN-ZOOM galaxies which tend to remain more disordered and bursty, potentially facilitating more efficient LyC escape. A detailed investigation of the physical origin of these differences is

beyond the scope of this work and will be explored in a forthcoming study(Puchwein et al., in prep.).

We have also adapted our statement that ``high UV luminosity correlates with a high escape fraction" and have softened our discussion of the f_esc - M_uv relation in Section 3.1. We now make clear that our argument refers to the population-level median trend shown in Fig. 1 which does not imply a unique proportionality between M_uv and f_esc. The higher scatter at the faint end permits individual UV-dim galaxies to exhibit very high escape fractions, including the highest values in our sample; this does not contradict the reported trend, which we remain confident is representative of our THESAN-ZOOM sample.

Finally, while the number of galaxy realisations decreases towards the high-mass end, we do not expect this to significantly impact the reported trend as our sample contains 1637 galaxy realisations with stellar masses greater than10^9 M_⊙, ensuring that this regime remains well sampled.

We have revised the text in section 3.1 accordingly and broadened the literature discussion to include the studies suggested by the referee.

- Page 6 line 56L: The information content of the UV slope is not only limited to tracing dust attenuation. It also carries information about the age and ionizing properties of the stellar population as well as the state of the ISM (through the nebular continuum).

We have expanded our discussion about the UV slope in Section 3.2 to clarify that it carries information on the stellar populations and the state of the ISM, as well as the UV dust attenuation.

- Have the authors explored the use of stellar age (e.g. ionizing-luminosity weighted) as a tracer? Alternatively, given the high importance of SFR10/SFR100 (as an indicator of overall SFR burstiness), how would the introduction of other timescales (e.g. SFR3/SFR10) improve the model? These are potentially interesting properties which may carry information about whether sufficient time has passed after a burst for feedback to effectively clear channels for ionizing radiation.

We thank the referee for these suggestions. We have explored additional SFR timescales but found them to be insignificant when included in our models, as they are largely degenerate with the information already captured by SFR_10/SFR_100. Similarly, we expect an ionizing-luminosity-weighted stellar age to encode closely related information about the star-formation history, and therefore do not expect it to provide substantial additional predictive power over SFR_10/SFR_100. We now clarify this is Section 3.4.1.

- One of the key differences between THESAN-ZOOM and other RT-coupled simulations with a multi-phase ISM (e.g., SPHINX, MEGATRON) is the inclusion of the additional `early

feedback channel'. It would be important for the authors to discuss the impact that this might have on the physics of LyC escape (see e.g. Menon+2024 for a potentially useful reference). For example relevant to the present work, does the early feedback reduce the number of remnant leakers, therefore strengthening the connection between SFR-related indicators and Nion? This may also be interesting to explore in the context of the position of leakers on the main sequence, given the use of Delta MS (Figure A1, see also Figure 6 of Katz+2023)

We thank the referee for this suggestion. The early stellar feedback (ESF) channel in THESAN-ZOOM injects momentum into the gas around each newly formed star particle at a rate of 1000 km s-1 Myr-1 per unit stellar mass for the first 5 Myr after it forms (Kannan et al. 2025). It represents pre-supernova feedback that is unresolved or not explicitly modelled, and it makes star formation less clustered and less bursty than in runs without it (Kannan et al. 2025). We have added this description to Section 2.1.

THESAN-ZOOM retains a large population of remnant leakers despite the ESF. f_esc is anti-correlated with SFR_10/SFR_100 and is highest for SFR_10/SFR_100 <= 0.1 (at least ~10 Myr after the end of a burst; Fig. A1), by which time the ESF from the stars formed in the burst has ended. Likewise, f_esc falls by about an order of magnitude from ΔMS_10 ~ −1 to ΔMS_10 ~ 0 and is roughly flat above the main sequence (Fig. A1), so the strongest leakers lie below the main sequence, in the post-burst phase of the burst-quench cycle of THESAN-ZOOM galaxies (McClymont et al. 2025b). This is where remnant leakers, whose star formation is declining, are expected to lie (Katz et al. 2023a).

Second, the strong connection between SFR indicators and N_ion,esc (Fig. 3) mainly reflects the intrinsic ionizing photon production. The median N_ion,esc rises by ~4.5 dex between MUV = -13 and -19, of which only ~1.3 dex comes from the change in median fesc (Fig. 1). This connection therefore does not require LyC escape to coincide with the burst. If escape closely tracked ongoing star formation, SFR_10 would be expected to carry more information about N_ion,esc than SFR_100; instead, SFR_100 is the more important feature in Models B and D, consistent with escape that is delayed relative to the burst. We have added a discussion of these points to Section 3.4.

- Page 11 line 33L, \citealt{} should be used inside the brackets.

This has been implemented as advised.

- Given the intriguing and well-presented result that Q_HII evolution is strongly determined by scatter in the Nion-Muv relation, alongside the fact that this scatter is linked to the underlying model used, it would be useful to understand potential bias here. For example, if the RF model used is only trained on a subset of galaxies (e.g., Muv < -15, where the top-right panel of Figure 1 shows a slight kink), how would the subsequent analysis change?

We thank the referee for raising this point. We clarify that the scatter adopted in our reionization analysis, sigma = 0.725 dex, is measured directly from the residuals of the intrinsic THESAN-ZOOM relation and is not inferred from the RF predictions. Since both the relation and its scatter are derived from the same THESAN-ZOOM population, we do not expect the adopted scatter to be biased by, or otherwise influenced by, the RF analysis. Moreover, the strong dependence of the RF predictions on M_uv suppresses the intrinsic scatter of the underlying distribution, preventing an independent estimate of its value from the RF analysis. Instead, Section 4.5 addresses the possibility that the intrinsic scatter in THESAN-ZOOM may not be representative of the physical galaxy population. By varying sigma over a broad range of values, we explicitly quantify how such uncertainty affects the inferred reionization history.

- The running median of Figure C2 should be extended through to the bright galaxy end. While it is understood that the training set is sparse in this range, the behaviour of the model here is crucial to understand given the central conclusion that the late reionization is driven by bright sources. Furthermore, there appears to be an overall bias in the model of ~0.1-0.2 dex across the full range of Muv. How does this affect the reionization history once integrated across the full UVLF?

We thank the referee for this suggestion. We have revised Fig. C2 to extend the running median towards the bright end by calculating it in fixed-width M_uv bins. We retain bins with equal numbers of galaxies for the 16th - 84th percentile range, as these provide a more robust representation of the variation in the intrinsic scatter. The extended median also clarifies the apparent offset noted by the referee: while the median residual lies slightly above zero over the dimmer part of the M_uv range, the opposite becomes true towards the bright end. This indicates no evidence for an overall systematic bias in our N_ionesc - M_uv relation. The relation is obtained through an ODR fit to the same THESAN-ZOOM population as plotted, thus the small variations in the median residual primarily reflect slight departures of the underlying Nion,esc − MUV relation from the adopted linear parametrization, rather than an overall offset. Consequently, we do not find evidence for an overall systematic bias that could affect our reionization analysis.